

A Specification of an Anonymous Credential System Using BBS+ Signatures with Privacy-Preserving Revocation and Device Binding

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Abstract. Recently, there has been a growing interest in anonymous credentials (ACs) as they can mitigate the risk of personal data being processed by untrusted actors without consent and beyond the user's control. Furthermore, due to the privacy-by-design paradigm of ACs, they can prove possession of personal attributes, such as an authenticated government document containing sensitive personal information, while preserving the privacy of the individual by not actually revealing the data. Typically, AC specifications consider the privacy of individuals during the presentation of an AC, but often neglect privacy-preserving approaches for enhanced security features such as AC non-duplication or AC revocation. To achieve more privacy-friendly enhanced security features of non-duplication and privacy-preserving revocation, an AC can be partially stored on secure, trusted hardware and linked to a status credential that reflects its revocation status. In this paper, we specify an AC system that satisfies the requirements of minimality of information, unlinkability, non-duplication, and privacy-preserving revocation. This is achieved by adapting the hardware binding method of the Direct Anonymous Attestation protocol with the BBS+ short group signatures of Camenisch et al. and combining it with status credentials.

Keywords: anonymous credentials; privacy-preserving revocation; device binding; bbs+ signature

1 Introduction

The European Commission is committed to ensuring that all EU citizens have access to a digital identity by 2030, enabling them to use public and private sector online services while preserving their privacy[Eur21]. In many private and government applications, the abstract concept of digital identity is already in use for authentication and authorization purposes. Identity providers based on OpenID Connect [Ope], such as Google, enable single sign-on authentication for end users in the private sector [Goo]. Similarly, there are common public sector use cases, such as verifying individuals using the privacy-enhanced eIDAS notified German eID, based on Extended Access Control[BSI]. All in all, there are several approaches to implement digital identities, depending on the security and privacy guarantees required by the use case [PZ23, CVH02, MDNDD15, Hyp].

The aforementioned digital identity solutions are just a few examples of implementing digital identities using more or less different digital identity paradigms. Furthermore, it is the digital identity paradigm that determines the suitability of a digital identity solution for specific authentication and/or authorization purposes. For example, to comply with the strict requirements of the General Data Protection Regulation (GDPR) on privacy and data protection, a proficient digital identity paradigm is crucial. A particularly versatile paradigm, which is also referenced in the European Digital Identity Wallet Architecture and Reference Framework [Eur] (section 5.2

Available standardised formats) as one of the configuration requirements for the data model, is the Verifiable Credentials Data Model (VCDM) [SLC].

VCDM uses Verifiable Credentials (VCs) to implement digital identities. VCs are cryptographically signed claims made by an entity. Using VCs involves four interacting main actors: the issuers, who issue VCs; the holders, who hold VCs; the verifiers, who verify VCs; and the verifiable data registry that manages different types of metadata regarding VCs. Figure 1 illustrates fundamental interactions between the main actors, the issuing (1) and the presentation protocol (2). Specifically, during the presentation protocol, the holder sends the verifier only a verifiable presentation (VP) of its VC while typically not revealing the VC itself. Furthermore, Figure 1 clarifies the exchange of metadata, managed by the verifiable data registry, required to run the protocols. To clarify, in this paper, we define the verifiable data registry as a collection of systems that manage various types of metadata, such as system parameters or public keys. The simplicity of VCs makes them suitable for various authentication and authorization purposes, e.g., digital ID cards or tickets for public transportation.

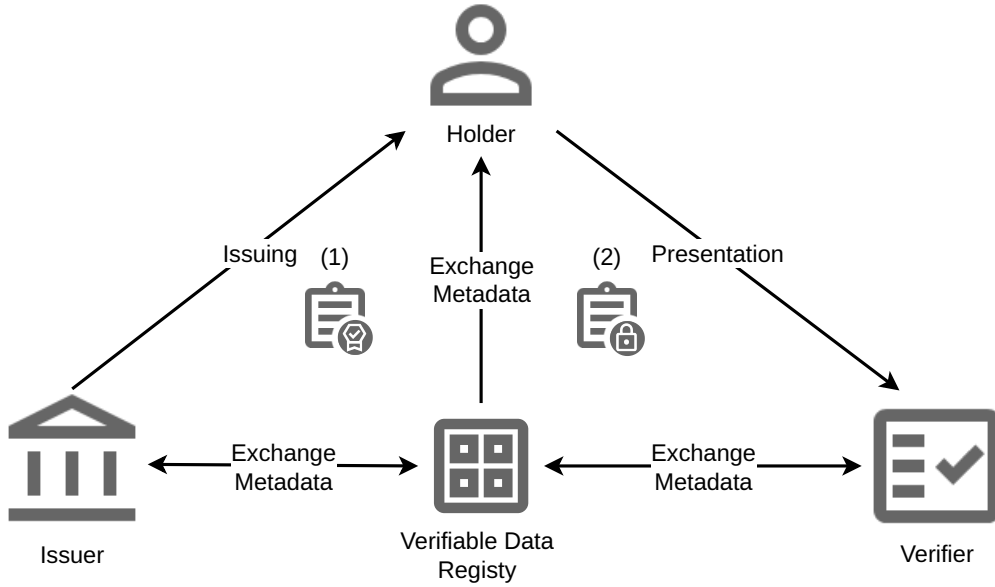


Figure 1: Schematic Illustration of a VC System.

Using VCs in a concrete use case, e.g., digital ID cards, imposes usability, privacy, and security requirements on VCs and their VPs. Richter et al. [RBSM23] discuss that beyond the obligatory security requirements for VCs (completeness, soundness, freshness, and unforgeability), there are challenging combinations of security requirements concerning holders' privacy. Furthermore, Bertram et al. [BRM24] also address security requirements in credential systems, focusing on third-party deniability, while emphasizing the challenges of hindering verifiers from proving the authenticity of VPs to a third party. However, in this paper, we specifically focus on minimality of information, unlinkability, non-duplication, and privacy-preserving revocation.

In the following, we define the four requirements, minimality of information, unlinkability, non-duplication, and revocation, which form a challenging combination.

Minimality of Information The VP must reveal only the information necessary for verification.

Unlinkability Even if a network of malicious verifiers cooperates without the holder’s consent, correlating two VPs must be practically impossible.

Non-duplication A VC cannot be duplicated, which means that it is effectively impossible to copy a VC.

Revocation VCs can be revoked. Revoking a VC makes it impossible to create a valid VP for it. The revocation of VCs may need some time to take effect but must be completed in a pre-defined timeframe.

Typically, in the literature [HS21, KMPQ23], VCs with VPs that satisfy minimality of information and provide strong holder authentication are referred to as ACs. If the VPs of an AC also satisfy unlinkability, the AC is referred to as a multi-show variant [CLT11]. In this paper, we aim to satisfy minimality of information and unlinkability, and thus, we always imply the multi-show variant when ACs are considered. By using appropriate signature schemes and Zero Knowledge Proofs (ZKPs), the combination of minimality of information and unlinkability is feasible [BBS04, CG08, ASMC13, LMPY16, PS16, CL04].

Next, non-duplication is usually achieved by storing the key material necessary to create a VP on a carrier medium (CM) in such a way that the previously defined target group cannot extract the key material. This strategy is commonly termed device binding. If the key material is not securely stored on a device, malicious holders could copy their AC and distribute it to others, which would be unacceptable in many use cases, e.g., when proving a person’s identity. Furthermore, employing ACs in digital identities poses a risk of identity theft. Specifically, a malware-infected carrier medium could allow an attacker to extract the AC and its cryptographic keys. To mitigate the risk of violating the requirement of non-duplication, it is best practice to use CMs with enhanced security features such as Secure Elements (SE) or Trusted Platform Modules (TPM).

If the non-duplication protection can be bypassed or access to the AC is impaired, such as loss of the carrier medium, there must be a way to revoke the AC. A common way to revoke a digital identity in conventional identity systems is to use a revocation state and a revocation list, where entries in a revocation list are identifiers associated with the revoked digital identities. In the case of a revocation list, a verifier checking the VP of an AC would use some kind of unique revocation identifier of the AC and check it against the entries of the revocation list. A unique and disclosed revocation identifier would violate the unlinkability requirement. Consequently, the challenge regarding revocation is to use an appropriate revocation mechanism (privacy-preserving revocation) that does not violate the unlinkability requirement. Fortunately, there are approaches to realizing a privacy-preserving revocation mechanism [RLW⁺22, HS21, Nyb96, CL02, VB22]. Nonetheless, due to its simplicity, in this paper, we propose to use the privacy-preserving revocation mechanism named Linked Validity Verifiable Credential (LVVC) by Freitag [Fre22].

Given the details of the requirements, minimality of information, unlinkability, non-duplication, and privacy-preserving revocation, it is clear that they limit the choice of cryptographic methods used for ACs, especially the method for signing and creating VPs. Recently, there has been growing interest in a suitable method, namely the BBS+ signature scheme as proposed by Au et al. [ASMC13]. Works using BBS+ signatures to prove the possession of ACs are the direct anonymous attestation protocol DAA+ by Camenisch et al. [CDL16a] and the AC system by Hesse et al. [HSS23]. However, the DAA+ protocol by Camenisch et al. [CDL16a] and the AC system by Hesse et al. [HSS23] only satisfy minimality of information, unlinkability, and non-duplication but do not comply with the requirement of privacy-preserving revocation. Since, to our knowledge, no research article specifies an AC system that uses BBS+ signatures and meets the requirement privacy-preserving revocation alongside to the requirements of minimality of information, unlinkability, and non-duplication, we propose a solution to close this gap in this paper.

1.1 Contribution

In this paper, we aim to describe a privacy-by-design AC system that satisfies minimality of information, unlinkability, non-duplication, and privacy-preserving revocation using BBS+. In the following, we summarize the contribution of this paper:

- Using some parts of DAA+, we specify an AC system enabling non-duplication to prevent identity theft.
- We replace the revocation mechanism of DAA+ with a time-based approach using LVVC to achieve actual privacy-preserving revocation.
- We discuss how the proposed AC system complies with the minimality of information, unlinkability, non-duplication, and privacy-preserving revocation requirements.

In summary, by combining established methods from DAA+ with LVVCs, our contribution describes an AC system that offers advanced privacy and security features.

1.2 Organization

In Section 2, we begin by discussing the methods used to achieve the results presented in Section 3, while providing essential definitions and foundations. In particular, Section 2.4 concludes Section 2 by giving a comparative overview of several AC systems regarding their compliance with minimality of information, unlinkability, non-duplication, and privacy-preserving revocation. Section 3 presents our proposed AC system by first giving a high-level overview followed by a specification of our construction using a bottom-up approach, beginning with low-level functionalities in Sections 3.1 and 3.2 and ending with more complex protocols in Section 3.3. In addition, Section 3.4 addresses the compliance of the proposed AC system with the requirements of minimality of information, unlinkability, non-duplication, and privacy-preserving revocation. Finally, Section 4 discusses the results presented in Section 3 and ends with a conclusion.

2 Materials and Methods

The following sections describe the selection process of methods that we use to specify the proposed AC system. In the selection process, we focus on the requirements of minimality of information, unlinkability, non-duplication, and privacy-preserving revocation. Furthermore, especially for the selected methods to achieve minimality of information, unlinkability, and privacy-preserving revocation, extensive foundations are necessary, which are introduced as subsections to the respective requirements. The materials and methods section concludes with an overview of various AC systems, which are compared with each other in a table regarding their compliance with the requirements for method selection.

2.1 Minimality of Information and Unlinkability

Several suitable signature schemes for AC systems satisfy the minimality of information and unlinkability requirements [LMPY16, PS16, CL04]. One of the oldest and best-known signature schemes for ACs is the Camenisch and Lysyanskaya (CL) signature [CL03] in a slightly modified version by Camenisch and Groß [CG08]. However, signature schemes such as the aggregatable attribute-based equivalence class signature (AAEQ) presented by Hanzlik and Slamanig [HS21] or the BBS+ signature by Au et al. [ASMC13], which is a modification of the BBS scheme developed by Boneh, Boyen, and Shacham [BBS04] are worth mentioning. Further, Tessaro and Zhu [TZ23] presented that BBS offers an 891-bit shorter signature than BBS+ with the same security guarantees, which makes BBS an excellent candidate for an AC system. Nevertheless, the

European Digital Identity Wallet Architecture and Reference Framework [Eur] specifies VCDM v2.0 and VCDM v1.1 for electronic attestation. Contrary to VCDM v2.0, VCDM v1.1 does not mention BBS for signatures. In fact, VCDM v1.1 references only one cryptographic suite suitable for AC systems that use BBS+ signatures. Hence, in this paper, we utilize BBS+ signatures.

2.1.1 BBS+ Signature Scheme

For the formal description of the AC system in Section 3, it is important to establish some basic concepts, such as groups with a bilinear map, special notation, and the BBS+ signature scheme and its underlying security assumption.

Pairing-based cryptography using cyclic groups is used extensively in Section 3. Therefore, $\mathbb{G}_1$, $\mathbb{G}_2$, and $\mathbb{G}_T$ denote cyclic groups of prime order p , $\mathbb{Z}_p$ a finite field (integer modulo p) over prime p and $1_{\mathbb{G}}$ the identity element in the group $\mathbb{G}$.

Definition 1. (*Pairing [GPS08]*) An efficient evaluable map $e : \mathbb{G}_1 \times \mathbb{G}_2 \mapsto \mathbb{G}_T$ is called a pairing if it satisfies bilinearity and non-degeneracy, i.e., for all generators $g_1, g'_1 \in \mathbb{G}_1, g_2, g'_2 \in \mathbb{G}_2$ the equations $e(g_1 \cdot g'_1, g_2) = e(g_1, g_2) \cdot e(g'_1, g_2)$, $e(g_1, g_2 \cdot g'_2) = e(g_1, g_2) \cdot e(g_1, g'_2)$, and $e(g_1, g_2) \neq 1_{\mathbb{G}_T}$ hold.

Next, based on Definition 1, we define a structure termed as *bilinear group with generators* (BG) in Definition 2.

Definition 2. (*Bilinear Group with Generators*) A bilinear group with generators, generated by a Probabilistic Polynomial Time (PPT) algorithm $\mathcal{G}$ with security parameter 1^λ as input, is defined as $\text{BG} = (\mathbb{G}_1, \mathbb{G}_2, \mathbb{G}_T, e, p, g_1, g_2, g_T)$ where $g_1 \in \mathbb{G}_1$, $g_2 \in \mathbb{G}_2$, and $g_T \in \mathbb{G}_T$ are generators and $e : \mathbb{G}_1 \times \mathbb{G}_2 \mapsto \mathbb{G}_T$ is a pairing.

To avoid introducing generators in later contexts, Definition 2 explicitly mentions generators in BG. Additionally, in this paper, when using BG, we only consider type-3 pairings, which means that $\mathbb{G}_1 \neq \mathbb{G}_2$ and there is no efficient isomorphism $\psi : \mathbb{G}_2 \mapsto \mathbb{G}_1$ (see Galbraith et al. [GPS08] for more information). Furthermore, the security parameter 1^λ , which is used to generate a BG, defines the level of security in bits.

For applying pairings and BGs in a cryptographic context, in this paper, we use the following two notations. First, $x \xleftarrow{\$} S$ means that x is drawn uniformly at random from a set S . Second, throughout this paper, we specify algorithms using the notation:

bold name of the algorithm : input $\mapsto$ output: followed by a description of the steps that generate the output.

With the foundations established previously, it is now possible to explore the application of Definition 1 and Definition 2 in a practical setting, such as the BBS+ signature scheme of Au et al. [ASMC13] defined in Figure 2.

Regarding the security of the BBS+ signature scheme, Camenisch et al. [CDL16a] prove the BBS+ signature scheme secure under the q -Strong Diffie-Hellman assumption (q -SDH) for type-3 pairings. The q -SDH assumption is originated by Boneh and Boyen [BB08] and Definition 3 is a compact representation of the asymptotic formulation of the q -SDH assumption for the q -SDH problem.

Definition 3. (*q -SDH Assumption*) Let BG be a bilinear group with generators. The q -SDH assumption holds for BG if, for every PPT algorithm $\mathcal{A}$,

$$\text{Adv}_{q, \mathcal{A}}(\lambda) := \Pr \left[\mathcal{A}(g_1, g_1^x, \dots, g_1^{(x^q)}, g_2, g_2^x) = \left(c, g_1^{\frac{1}{x+c}} \right) \right]$$

is a negligible function in λ where $q : \mathbb{Z} \mapsto \mathbb{Z}$ is a polynomially bounded function of λ .

Another notable foundation is the signature proof of knowledge (SPK) of discrete logarithms and statements about them as defined in [CS97], which uses the Fiat-Shamir heuristic [FS87] in

Parameter Description: • Bilinear group with generators BG • Security parameter 1^λ • Number of messages $L \geq 1$ • Message $M = (m_1, \dots, m_L) \in \mathbb{Z}_p^L$ • Secret key $x \in \mathbb{Z}_p^*$ • Public key $pk = (w, h_0, \dots, h_L) \in \mathbb{G}_2 \times \mathbb{G}_1^{L+1}$ where $w = g_2^x$ • BBS+ signature $\sigma = (A, e, s) \in \mathbb{G}_1 \times \mathbb{Z}_p^2$
Setup : $(1^\lambda) \mapsto \text{BG}$: Generate a bilinear group with generators BG according to 1^λ . KeyGen : $(\text{BG}, L) \mapsto (pk, x)$: 1. $(h_0, \dots, h_L) \xleftarrow{\\$} \mathbb{G}_1^{L+1}$ 2. $x \xleftarrow{\\$} \mathbb{Z}_p^*$ 3. $w \leftarrow g_2^x$ 4. $pk \leftarrow (w, h_0, \dots, h_L)$ Sign : $(\text{BG}, M, pk, x) \mapsto \sigma$: 1. $e, s \xleftarrow{\\$} \mathbb{Z}_p$ 2. $A \leftarrow (g_1 \cdot h_0^s \cdot \prod_{i=1}^L h_i^{m_i})^{\frac{1}{e+s}}$ 3. $\sigma \leftarrow (A, e, s)$ Verify : $(\text{BG}, pk, M, \sigma) \mapsto f$: If $e(A, w \cdot g_2^e) = e(g_1 \cdot h_0^s \cdot \prod_{i=1}^L h_i^{m_i}, g_2)$ set $f \leftarrow 1$ otherwise $f \leftarrow 0$.

Figure 2: The specification of BBS+ signatures.

the random oracle model [BR93]. The following example highlights the notation used for SPK. Let $m \in \{0, 1\}^\lambda$ be a message, $\alpha \in \mathbb{Z}_p^*$ a witness, $g_1 \in \mathbb{G}_1$ a basis, and $y = g_1^x$ a statement, then $\text{SPK}\{(\alpha) : y = g_1^x\}(m)$ denotes a signature proof of knowledge of α such that $y = g_1^x$ using m in the Fiat-Shamir heuristic. Since all steps to generate the SPKs used in Section 3 are specified in Section 3.2, we neglect to define the SPK notation formally here.

2.2 Non-duplication

For AC systems, non-duplication protection is not a new idea and has been studied in several academic papers for different signature schemes and mainly two architectural concepts. First, there are AC systems that basically implement all operations of the signature scheme in a CM, such as an SE [BCGS09, BHJ⁺10]. Implementing the signature scheme in a CM has the advantage that it is particularly robust against secret key extraction attacks. However, the disadvantage of this naive approach is that in addition to the key material, all attribute data must also be stored in the CM. Usually, due to the limited resources of CMs (e.g., SEs), storing all AC data in a CM is hardly an option, especially in use cases with several ACs or numerous attribute data. Second, as an alternative to the naive non-duplication protection approach, there is the idea of splitting the computations between the CM (core) and the host device (helper). On the one hand, splitting the computations results in a higher risk for successful key extraction due to the more complex architecture, on the other hand, resource weaknesses of the CM can be

compensated by the host device. After explaining the main advantages and disadvantages of the two types of non-duplication protection, the following paragraphs discuss papers that implement the core/helper design.

Hanzlik and Slamanig [HS21] introduce the core/helper approach and utilizing aggregatable attribute-based equivalence class signatures. Their specification involves a core device with limited resources (e.g., an SE) and a helper device with relatively more computational resources (e.g., a smartphone). In particular, Hanzlik and Slamanig implement their AC system using a standalone NFC-based Multos SE as the core.

The anonymous credential system by Hesse et al. [HSS23] also uses the core/helper approach. Hesse et al. propose the use of a standalone smart card as the carrier medium and a smartphone as the helper device, but instead of aggregatable attribute-based equivalence class signatures, they use the BBS+ signature scheme. In addition, for on-site use cases, Hesse et al. propose visual authentication of a holder by a person checking the holder's photo printed on the smart card.

Another solution that matches the idea of the core/helper approach is DAA+ by Camenisch et al. [CDL16a]. However, contrary to the work of Hesse et al., DAA+ uses a TPM 2.0 as the core [Trub]. In fact, DAA+ is not an AC system, but rather a protocol for privacy-preserving remote attestation. Nevertheless, DAA+ has many features that are necessary to meet the requirements that we state for an AC system, such as the use of ZKPs, which can be used to prove the possession of ACs. Hence, in this paper, we interpret and extend DAA+ towards an AC system.

The approach we use in this paper is similar to that of Camenisch [Cam06] using the Direct Anonymous Attestation (DAA) protocol as a basis for specifying an AC system. Unlike our approach, Camenisch considers the DAA protocol of the TPM V1.2 specification [Trua], which uses CL signatures to sign ACs and TPMs as core devices. Another difference from our work is that Camenisch suggests that holder can use their ACs on multiple devices.

2.3 Privacy-preserving Revocation

From the aforementioned third-party research on AC systems using a core/helper approach, only the solution by Hanzlik and Slamanig [HS21] briefly discusses a solution for privacy-preserving revocation using a revocation list with randomized public device keys pk_i^R . Hanzlik and Slamanig propose that each time an AC is presented to a verifier, a holder must generate ZKPs of inequality for all pk_i^R that are also linked in VPs of a valid AC. The revocation list approach increases the runtime of the presentation protocol linearly by the number of pk_i^R , and thus is not adopted in this article.

Apart from the revocation list, there are several possible approaches to implementing privacy-preserving revocation [LKDDN11, Fre22], one of the most popular being the cryptographic accumulator [RLW⁺22]. In simple terms, cryptographic accumulators are used in the context of AC systems to compress information about the revocation state of an AC into mathematical structures, so that a holder can prove the knowledge of a mostly signed piece of information that is or is not part of the accumulator, which in turn determines whether an AC is revoked or not. Common constructions for accumulators are based on the RSA assumption [Nyb96], the Strong RSA assumption [CL02], or the Strong Diffie-Hellman assumption [VB22].

Regarding the AC system we propose in this paper, we see that the main problem with the accumulator approach is that it would increase the complexity of the system. In particular, an entity would have to be designated to manage the accumulator, and the ZKP to prove possession of an AC would have to be significantly expanded. Therefore, we do not use accumulators in this work.

To our knowledge, Freitag [Fre22] was the first to mention the idea of LVVCs scientifically and to compare it to other, more or less privacy-preserving revocation mechanisms. In particular, LVVCs are VCs, and in an AC system that uses LVVCs, the LVVCs contain information about the revocation status of an AC (see Section 2.3.1 for more details). To prove the revocation status

of an AC, a holder must link the VPs of that AC and an LVVC. Due to its simplicity, the AC system we present in this paper uses an LVVC-based revocation mechanism.

2.3.1 Linked Validity Verifiable Credentials (LVVCs)

Freitag [Fre22] proposes LVVCs as privacy-preserving revocation mechanism for AC systems. LVVCs are ACs with only the information about the revocation status, named revocation information, of a corresponding AC. Since LVVCs are ACs, the same cryptographic primitives are used to sign their claims.

Figure 3 schematically illustrates how an AC system can use LVVCs to realize its revocation mechanism. Using an AC in conjunction with an LVVC requires three steps. First (1), the issuer must issue an AC to the holder. Second (2), before a holder can use its AC, the issuer must issue an LVVC containing the revocation information of the holder’s AC. Third (3), the holder creates linkable VPs for its AC and LVVC and presents the result to the verifier for verifying the VPs and their link. The verifier accepts the AC only if both VPs are valid, the VPs are linked, and the VP of the LVVC contains proper revocation information.

As step (2) in Figure 3 indicates, the issuer manages the revocation status of ACs. Depending on the revocation information, the LVVC might be valid only temporarily. In the case of short-lived LVVCs, it can be necessary for the holder to get valid LVVCs for an AC regularly. To clarify, theoretically, as long as an AC is valid, the holder can obtain a corresponding LVVC from the issuer.

When using LVVCs as a revocation mechanism in an AC system, it is crucial to implement them in compliance with their privacy requirements. The revocation information and the link between VPs must not compromise the holder’s privacy. Therefore, the following two paragraphs highlight how the AC system described in Section 3 utilizes LVVC as a privacy-preserving revocation mechanism.

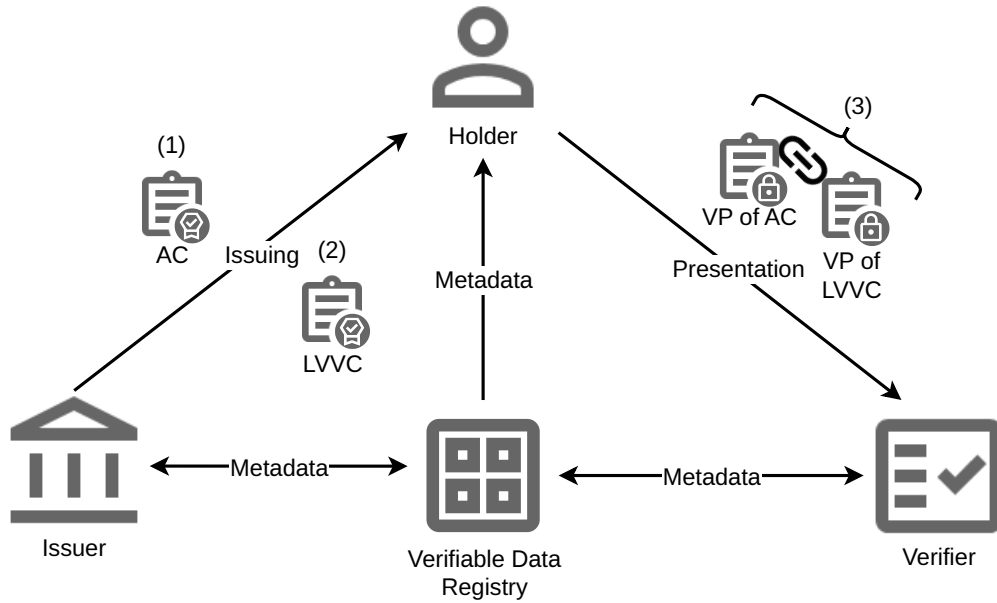


Figure 3: Schematic Illustration of an AC System Using LVVCs.

As Section 3 describes an AC system with a privacy-preserving revocation mechanism using

LVVCs, it is necessary to specify how to link the VPs of LVVC and AC in a privacy-preserving manner. Section 3 describes establishing a privacy-preserving link between LVVC and AC with ZKPs. However, in addition to privacy-preserving linking, the AC system must also choose a privacy-preserving revocation information.

Like Freitag [Fre22], we propose that the LVVC contains time-based revocation information. Section 3.4 discusses that a verifier can check the time-based revocation information of ACs in a privacy-preserving manner. In particular, a verifier only accepts ACs linked to LVVCs with an unexpired timestamp.

Using LVVCs with time-base revocation information introduces additional BBS+ signing overhead. From the benchmarks by Flamini et al. [FSS⁺24], we can conservatively estimate that generating a BBS+ signature takes approximately 2 ms on modern consumer hardware. Hence, using this estimation, issuing an LVVC would cost the issuer about 2 ms computation time. Nevertheless, to avoid a bottleneck on the issuer side, one can provide hardware that meets the demand. In addition, the issuer can precompute LVVCs to be ready when requested. Precomputing LVVCs has the potential to prevent signing calculations on the issuer side while issuing an LVVC, thus preventing delays.

Besides the performance of a time-based revocation mechanism with LVVC, more pros and cons regarding its usability should be considered. Time-based LVVCs do not allow active revoking of ACs, making them inapplicable in use cases where this is necessary. However, using time-based revocation information has the advantage that distributing revocation information to verifiers is unnecessary, making it applicable in scenarios where the verifier rarely has an internet connection. Moreover, it is noted that LVVCs may be solicited via a background process without the holder's involvement. Nevertheless, one disadvantage of issuing LVVCs is that the host device must be online to obtain an LVVC. Whether the requirement that a host device must be online for receiving LVVCs is acceptable must be decided depending on the use case. Ultimately, using time-based LVVCs is acceptable for many cases where active revocation is unnecessary while decreasing the complexity of the AC system.

2.4 Structured Comparison

Table 1 presents a systematic evaluation of various anonymous credential (AC) systems based on five criteria: minimality of information, unlinkability, non-duplication, revocation, and privacy-preserving revocation. The purpose of the comparison in Table 1 is to give the reader a rough overview of the current state of research to be able to adequately classify the specification of an AC system proposed in Section 3. It should be noted that the comparison in Table 1 does not claim to be exhaustive. For a more comprehensive overview of AC systems, see the SoK by Kakvi et al. [KMPQ23].

Table 1 highlights that all the compared systems achieve minimality of information. Various signature schemes were used by the AC systems in Table 1. The fact that BBS+ is among the methods used indicates its relevance in the scientific literature on AC systems and further justifies its suitability for the AC system we propose in this paper.

Unlike the minimality of information, not all AC systems in Table 1 satisfy unlinkability. Specifically, only U-Prove [PZ23, Paq23] does not meet unlinkability. In detail, U-Prove uses single-show ACs, which means that two or more VPs of an AC can be correlated. This means that U-Prove does not meet our definition of ACs. Additionally, also Kakvi et al. [KMPQ23] state that U-Prove is technically not an AC system. However, U-Prove is still included here because it is a relevant VC system that has privacy-preserving properties. Besides, all specifications displayed in Table 1 that comply with the unlinkability requirement use ZKPs.

The non-duplication strategies vary across the specifications in Table 1. Some AC systems in Table 1 adopt the core/helper approach, while others use CMs exclusively. Overall, the comparison reveals that employing appropriate hardware to fulfill the requirement of non-duplication is an established practice.

Subsequently, we explore revocation and privacy-preserving revocation. In particular, Hesse et al. [HSS23] and Batina et al. [BHJ⁺10] do not detail any revocation method for ACs. In contrast, Bichsel et al. [BCGS09] and U-Prove employ a revocation list that negatively affects the unlinkability of ACs upon revocation. As indicated in Table 1, only Hanzlik and Slamanig [HS21], along with Idemix [BC10], manage to integrate privacy-preserving revocation techniques.

Ultimately, the comparison in Table 1 underscores the strengths and limitations of existing AC systems. Table 1 emphasizes the absence of an AC system that utilizes BBS+ signatures while satisfying all the requirements of minimality of information, unlinkability, non-duplication, and privacy-preserving revocation. Hence, the AC system that we specify in Section 3, which uses BBS+ signatures and complies with the requirements of minimality of information, unlinkability, non-duplication, and privacy-preserving revocation, represents a novel contribution.

Table 1: Comparative evaluation of various AC system specifications.

Specification	Minimality of Information	Unlinkability	Non-Duplication	Revocation	Privacy-Preserving Revocation
Hanzlik & Slamanig [HS21]	AAEQ	AAEQ ZKPs	& Core/helper	Revocation list	Yes
Hesse et al. [HSS23]	BBS+	BBS+ ZKPs	& Core/helper	No	No
Bichsel et al. [BCGS09]	CL	CL & ZKPs	CM only	Revocation list	No
Batina et al. [BHJ ⁺ 10]	Self-blindable ECDSA	Self-blindable ECDSA & ZKPs	No	No	
U-Prove [PZ23, Paq23]	Blind digital signatures [ISO16]	No	CM only	Revocation list	No
Idemix [CL03, CL01, BC10]	CL	CL & ZKPs	CM only [VA13]	Cryptographic accumulator	Yes

3 Results

This section describes the algorithms and protocols that form the anonymous credential system Π_{bbs+} . The specification of algorithms and protocols of Π_{bbs+} , especially those concerning SPKs, is strongly based on DAA+ of Camenisch et al. [CDL16a]. In contrast to DAA+, the anonymous credential system Π_{bbs+} realizes revocation via LVVC. Therefore, Π_{bbs+} strips the revocation list from DAA+ and introduces a protocol to issue LVVCs. In addition, Π_{bbs+} renames and bundles some algorithms of DAA+ to fit in the context of anonymous credentials. Furthermore, unlike DAA+, we propose a broad framework that is not limited to TPMs but is compatible with other CMs offering similar security characteristics. We also point out that the CM, whether integrated into or separate from the host device, must support secure data exchange as outlined below. In short, we heavily rearrange algorithms, rename labels, and exchange components of DAA+ to specify Π_{bbs+} .

The AC system Π_{bbs+} consists of several algorithms and protocols. To present Π_{bbs+} as clearly

as possible, we provide three subsections, where Section 3.1 defines the algorithms necessary to use BBS+ signatures, Section 3.2 describes all SPKs used, and Section 3.3 defines high-level protocols that apply the algorithms introduced in Sections 3.1 and 3.2. Before presenting the algorithms and protocols of Π_{bbs+} , the following three paragraphs specify essential notation, simplifications, and conditions.

The actors in Π_{bbs+} are systems of the verifiable data registry, holder, verifier, and issuer ($\mathcal{I}$), where the holder has a host device (e.g., a smartphone) with an integrated or separated CM called $\mathcal{CM}$. In addition, $\mathcal{I}$ and $\mathcal{CM}$ denote unique identifiers, and the notation $\mathcal{CM}.\text{protocolname}()$ means that the CM identified by $\mathcal{CM}$ executes the protocol identified by protocolname . Further, we skip details of the issuer and verifier, as they depend on concrete implementations, which vary on the requirements of the underlying use case.

Another simplification in Π_{bbs+} is the use of ideal functionalities as defined in [Can01]. To be precise, the *common reference string functionality* $\mathcal{F}_{crs}$ is available to all parties and is used to distribute the required system parameters [CDL16b]. Furthermore, the *certificate authority functionality* $\mathcal{F}_{ca}$ is used to register and retrieve the issuer's public key [Can04]. Note that in Π_{bbs+} , the ideal functionalities $\mathcal{F}_{crs}$ and $\mathcal{F}_{ca}$ specify the systems of the verifiable data registry. Moreover, in Π_{bbs+} , the *secure communication functionality* $\mathcal{F}_{smt}^l$ models the secure message exchange between a host device and various entities, such as a CM, an issuer, and a verifier [CDL16b]. Since secure communication between the host device and the issuer is done using $\mathcal{F}_{smt}^l$, message exchange between a CM and an issuer only needs an authentic channel. To clarify, the authentic channel is realized by the *authenticated communication functionality* $\mathcal{F}_{auth*}$. For the sake of readability, the specification of Π_{bbs+} does not explicitly mention the use of $\mathcal{F}_{crs}$ and $\mathcal{F}_{smt}^l$.

Next, the following describes general assumptions and facts about the data used to execute Π_{bbs+} algorithms and protocols. When executing algorithms or protocols in Π_{bbs+} , it is assumed that actors always refer to the correct system parameters, such as the number of attributes L of an AC, the security parameter 1^λ , or the $\text{BG} = (\mathbb{G}_1, \mathbb{G}_2, \mathbb{G}_T, e, p, g_1, g_2, g_T)$ of an issuer's public BBS+ key pk . Some protocols of Π_{bbs+} work with persistent records, which are managed by store, load, and delete operations. To clarify, we neglect to specify the management operations in detail, since it is sufficient to know that these operations store, load, and delete records regarding an identifier. Furthermore, if any algorithm of Π_{bbs+} fails a check or management operation, the algorithm must report an appropriate **ExitCode** to all participants and stop immediately. Additionally, **ExitCode** and the validity of the key material of the issuer are also implicitly checked.

Figure 4 provides an informal high-level overview of Π_{bbs+} using the example of a smartphone as a host device with an integrated or separate carrier medium. The following briefly adds some details on the numbered protocol flows. For a formal specification on the numbered protocol flows, see Section 3.3.

The first protocol (1) that must be executed in Π_{bbs+} is the **IssuerSetup** that generates all the necessary data structures to issue ACs. Especially, **IssuerSetup** generates the BBS+ key material and transmits the public key pk to $\mathcal{F}_{ca}$. Furthermore, after running (1), $\mathcal{F}_{ca}$ distributes pk to the host device and the verifier when necessary.

After **IssuerSetup**, the issuer can issue ACs to a host device. The **Issuing** protocol (2) issues ACs realized by a BBS+ signature to a host device. In particular, during **Issuing**, the CM creates a secret device key that never leaves the CM. The CM then passes the public device key $Q = h^{sk}$ to the application running on the host device, which in turn forwards it to the issuer that incorporates Q while generating the signature. After successfully verifying that Q originates from a CM, the issuer generates the BBS+ signature and transmits the AC to the host device.

Next, in Π_{bbs+} , the host device must use the **LVVCIssuing** protocol (3) to request an LVVC that includes the revocation status of the AC. Under the condition that the AC is not revoked and the host device successfully authenticates itself against the issuer, it obtains an LVVC from the issuer. Like the AC, the LVVC is realized using a BBS+ signature. In addition, the issuer

incorporates Q while generating the BBS+ signature of the LVVC, ensuring that only a host device with a CM holding the device secret key sk can use the LVVC during presentation against a verifier. A host device holding an AC and a corresponding LVVC is ready to use them in the **Presentation** protocol (4).

Finally, based on all the prep work, the **Presentation** protocol (4) can prove some attributes of a holder's AC. To clarify, the verifier requests the host device to disclose a set of its AC and LVVC attributes and to prove them using their VPs. With the holder's consent, the host device creates the requested VPs and transmits them to the verifier. After receiving the VPs, the verifier checks the revocation status and the AC attributes. Based on the result of checking the VPs of the AC and LVVC, the verifier performs actions (e.g., granting access to a service), whose description is beyond the scope of this paper.

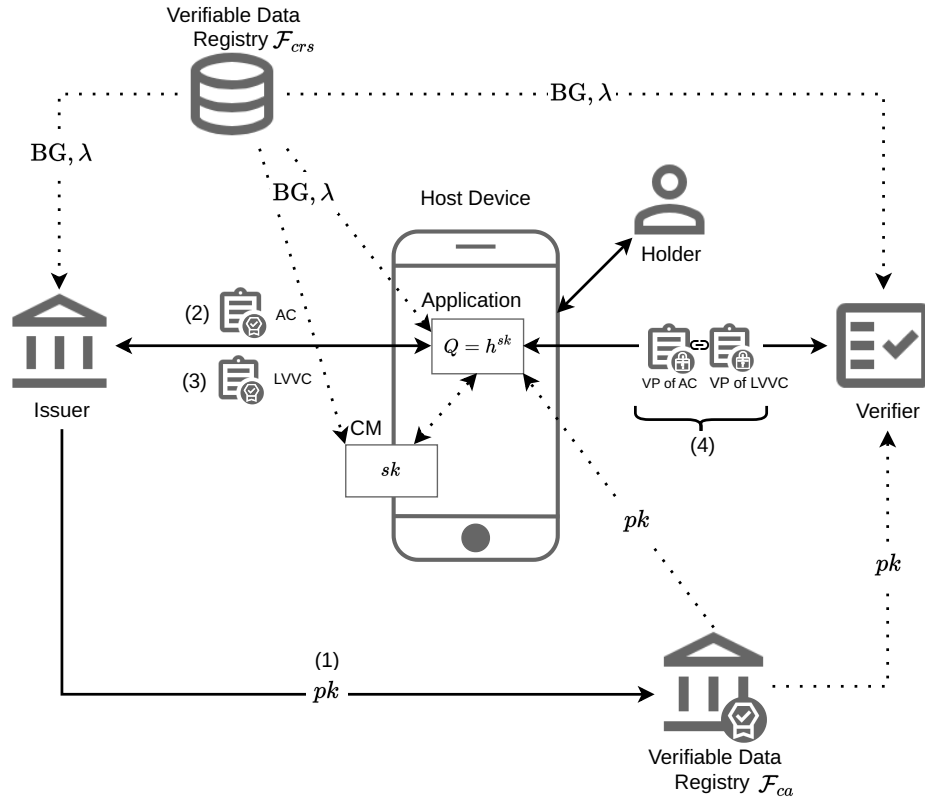


Figure 4: A schematic high-level system overview of Π_{bbs+} using the example of a smartphone as a host device with an integrated or separate carrier medium. The figure illustrates the protocol flows between the actors of Π_{bbs+} and the distribution of some parameters. The dotted arrows represent the parameter distribution, and the solid arrows symbolize interactions during protocol flows. Furthermore, pk denotes the issuer's public key, sk the secret device key, and $Q = h^{sk}$ the public device key.

3.1 Signature Scheme

At its core, Π_{bbs+} certifies ACs using BBS+ signatures, and this section specifies their use. The following specification of BBS+ signatures differs from the specification in Section 2.1.1, but is similar to that of Camenisch et al. [CDL16a]. Analogous to [CDL16a], the specification here uses a public device binding key $Q = h^{sk} \in \mathbb{G}_1$ with a private device binding key $sk \in \mathbb{Z}_p^*$ to generate a signature. However, unlike [CDL16a], the underlying generator h is described here without an index to differentiate it from other generators associated with attributes. Due to the deviations from [CDL16a] and to ensure clear protocol definitions, Figure 5 specifies the BBS+ scheme used in Π_{bbs+} .

Parameter Description: • Security parameter 1^λ • Number of attributes $L \geq 1$ • Attributes $attrs = (a_1, \dots, a_L) \in \mathbb{Z}_p^L$ • Secret key $x \in \mathbb{Z}_p^*$ where $sk \in \mathbb{Z}_p^*$ • Public key $pk = (w, h, h_0, \dots, h_L) \in \mathbb{G}_2 \times \mathbb{G}_1^{L+2}$ • Public device binding key $Q = h^{sk} \in \mathbb{G}_1$ • BBS+ signature $\sigma = (A, e, s) \in \mathbb{G}_1 \times \mathbb{Z}_p^2$
ISetup : $1^\lambda \mapsto \text{BG}$: According to 1^λ, generate a bilinear group with generators BG. IKGen : $L \mapsto (x, pk)$: 1. $(h, h_0, \dots, h_L) \xleftarrow{\\$} \mathbb{G}_1^{L+2}$ 2. $x \xleftarrow{\\$} \mathbb{Z}_p^*$ 3. $w \leftarrow g_2^x$ 4. $pk \leftarrow (w, h, h_0, \dots, h_L)$ IGenSig : $(pk, x, Q, attrs) \mapsto \sigma$: 1. $e, s \xleftarrow{\\$} \mathbb{Z}_p$ 2. $A \leftarrow (g_1 \cdot h_0^s \cdot Q \cdot \prod_{i=1}^L h_i^{a_i})^{1/(e+x)}$ 3. $\sigma \leftarrow (A, e, s)$ VerifySig : $(pk, \sigma, Q, attrs) \mapsto f$: 1. $b \leftarrow g_1 \cdot h_0^s \cdot Q \cdot \prod_{i=1}^L h_i^{a_i}$ 2. Check that $A \neq 1_{\mathbb{G}_1}$ and $e(A, w \cdot g_2^e) = e(b, g_2)$ 3. If all checks pass set $f \leftarrow 1$ otherwise $f \leftarrow 0$

Figure 5: Specification of the BBS+ scheme used in Π_{bbs+}

3.2 Signature Proofs of Knowledge

First, this section specifies protocols to generate the SPKs used in Π_{bbs+} , and second, presents a method to verify a link between two specific SPKs. In addition, this section briefly explains

how and where Π_{bbs+} uses the specified algorithms. For high-level protocols incorporating these algorithms, the reader is referred to Section 3.3.

Before the SPK-generating protocols can be specified, essential definitions and notation must be clarified. In Π_{bbs+} , some algorithms use a join session identifier $jsid$ or a session identifier sid chosen in high-level protocols to load and store records. Furthermore, when generating the SPKs specified in this section, the random oracles $H : \{0, 1\}^* \mapsto \mathbb{Z}_p^*$ and $H_1 : \{0, 1\}^* \mapsto \mathbb{G}_1$ are used, where all SPKs utilize H to implement the Fiat-Shamir heuristic, and some SPKs also apply H_1 to choose a generator in $\mathbb{G}_1$ for a given string.

Throughout this section, the names of the protocols that generate an SPK have the prefix “Gen”, and the names of the algorithms that verify an SPK have the prefix “Verify”. All SPK verification algorithms in Π_{bbs+} return 1 for a successful verification and 0 otherwise. In addition, the SPK verification algorithms share some input parameters. Therefore, this document displays their specification as a bundle at the end of Section 3.2. Unlike verifying SPKs, generating SPKs in Π_{bbs+} involves CMs, which makes SPK generation essential to illustrate how the non-duplication requirement is met.

The first SPK protocol specified in this section is **GenSPKQ**.

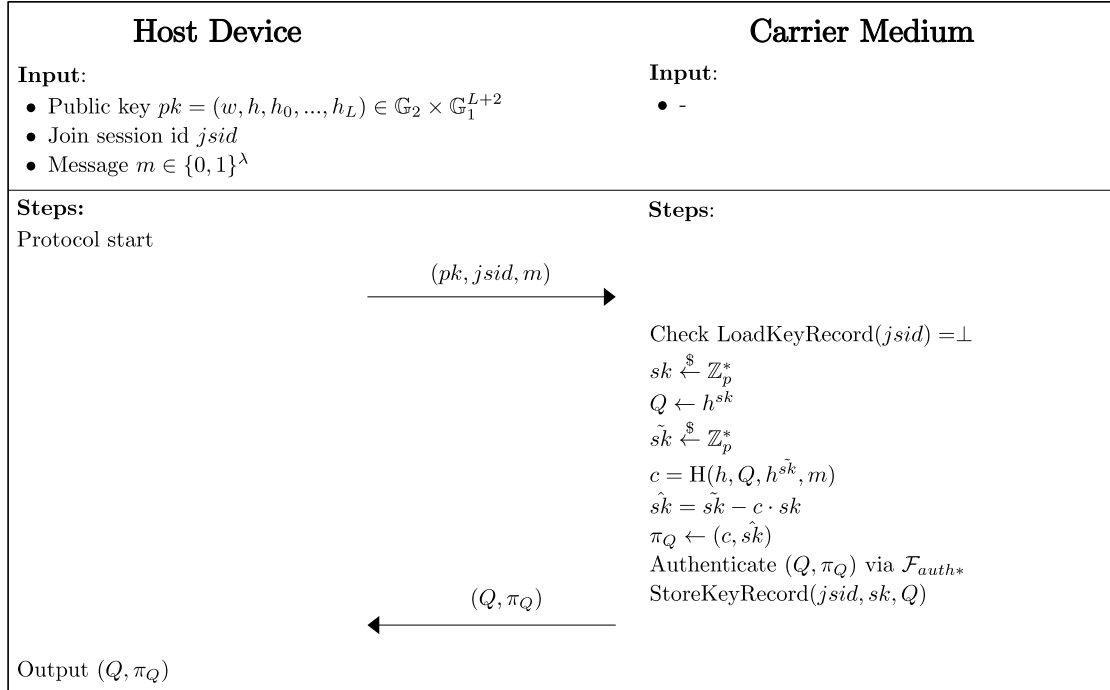


Figure 6: Specification of the **GenSPKQ** protocol.

As specified, **GenSPKQ** generates and stores a device secret sk and returns a public device key Q together with $\pi_Q \in (\mathbb{Z}_p^*)^2$ an SPK $\{(sk) : Q = h^{sk}\}(m)$ on message m . Because **GenSPKQ** generates and uses the device secret, the CM executes almost the entire protocol. In addition to generating an SPK, using $\mathcal{F}_{auth*}$ **GenSPKQ** proves that (Q, π_Q) originates from a CM. **GenSPKQ** is crucial in complying with the non-duplication requirement, as it builds trust that the ACs generated by an issuer using Q are bound to a CM (see the **Issuing** protocol in Section 3.3).

The proof of possession of an AC with **GenSPKSig** is the core of Π_{bbs+} . Specifically, a holder initiates the host device and the CM to execute **GenSPKSig** to generate the SPK. This SPK is used to prove possession of a valid BBS+ signature of an AC, and, therefore, of that AC. As

specified in Figure 7, for a set of undisclosed attributes $attrs_{\bar{D}} = \{(attrs)_j | j \in \bar{D}\}$, the algorithm **GenSPKSig** returns $\pi_\sigma = (A', \bar{A}, d, nym, \pi) \in \mathbb{G}_1^4 \times ((\mathbb{Z}_p^*)^7 \times \mathcal{P}(\mathbb{Z}_p^*))$ and

$$\begin{aligned} \text{SPK}\{(sk, attrs_{\bar{D}}, e, r_2, r_3, s') : \\ \bar{A}/d &= A'^{-e} \cdot h_0^{r_2} \\ \wedge g_1 \cdot \prod_{i \in D} h_i^{a_i} &= d^{r_3} \cdot h_0^{-s'} \cdot h^{-sk} \cdot \prod_{j \in \bar{D}} h_j^{-a_j} \\ \wedge nym &= H_1(bsn)^{sk}\}(m) \end{aligned}$$

on message m , where $\mathcal{P}(\mathbb{Z}_p^*)$ is the power set of $\mathbb{Z}_p^*$.

As mentioned above, possession of an AC is proven by an SPK with **GenSPKSig**, leaving the revocation status unresolved. To check the revocation status of an AC, Π_{bbs+} additionally uses LVVCs with revocation information. Like an AC, an LVVC has a BBS+ signature, the possession of which can be proven using **GenSPKSig**. To provide information about the revocation status, **GenSPKSig** must intentionally establish a link between the proofs of possession of an AC and an LVVC.

The **VerifyLink** algorithm, in Figure 8, specifies how to verify a link between two **GenSPKSig** proofs. In addition, it gives an idea of how to establish a link between two SPKs. Specifically, a review of Step 4 shows that using the same pseudonym nym during SPK generation establishes the link. In particular, using the same bsn and device secret for π_σ and π_σ^* leads to the link between nym and nym^* being the same since $nym = H_1(bsn)^{sk} = nym^*$.

Besides the **VerifyLink** algorithm, Figure 8 also specifies the SPK verification algorithms of Π_{bbs+} . As outlined, **VerifySPKQ** and **VerifySPKSig** perform verification on their respective SPKs. Given their sole focus on SPK validation, this document will not elaborate on them further.

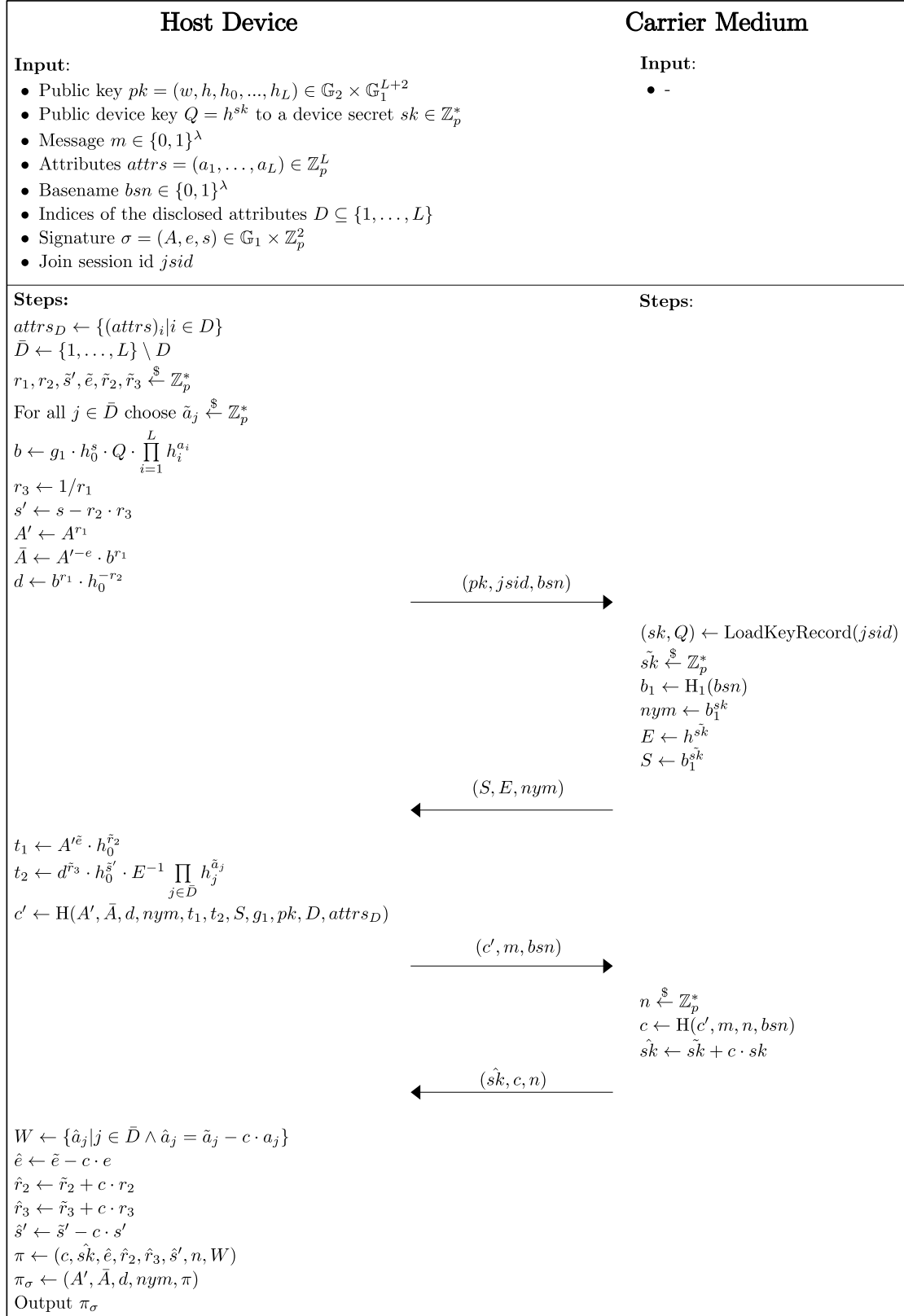


Figure 7: Specification of the GenSPKSig protocol.

Parameter Description:

- Public key $pk = (w, h, h_0, \dots, h_L) \in \mathbb{G}_2 \times \mathbb{G}_1^{L+2}$
- Public device key and SPK $(Q, \pi_Q) = (Q, c, \hat{sk}) \in \text{Im}(\text{GenSPKQ})$ where $Q = h^{sk}$ and $sk \in \mathbb{Z}_p^*$ is a device secret
- SPKs $\pi_\sigma, \pi_{\sigma^*} \in \text{Im}(\text{GenSPKSig})$ for BBS+ signatures σ and σ^*
- Messages $m, m^* \in \{0, 1\}^\lambda$
- Indices $D \subseteq \{1, \dots, L\}$ and $D^* \subseteq \{1, \dots, K\}$ of the disclosed attributes
- Sets of disclosed attributes $\text{attrs}_D = \{(\text{attrs})_i | i \in D\}$ and $\text{attrs}_{D^*} = \{(\text{attrs}^*)_j | j \in D^*\}$ where $\text{attrs} \in \mathbb{Z}_p^L$ and $\text{attrs}^* \in \mathbb{Z}_p^K$ are the attributes used to generate σ and σ^* respectively
- Basename $bsn \in \{0, 1\}^\lambda$
- Pseudonym $nym = H_1(bsn)^{sk} \in \mathbb{G}_1$ where $sk \in \mathbb{Z}_p^*$ is a device secret

VerifySPKQ : $(pk, Q, \pi_Q, m) \mapsto f$:

1. $t^* \leftarrow h^{\hat{sk}} \cdot Q^c$
2. $c^* \leftarrow H(h, Q, t^*, m)$
3. If $c^* = c$ set $f \leftarrow 1$ otherwise $f \leftarrow 0$

VerifySPKSig : $(D, \pi_\sigma, m, bsn, pk, \text{attrs}_D) \mapsto f$:

1. Check $A' \neq 1_{\mathbb{G}_1}$ and $e(A', w) = e(\bar{A}, g_2)$
2. $\hat{S} \leftarrow H_1(bsn)^{\hat{sk}} \cdot nym^{-c}$
3. $t_1^* \leftarrow A'^{\hat{e}} \cdot h_0^{\hat{r}_2} \cdot (\bar{A}/d)^{-c}$
4. $t_2^* \leftarrow (g_1 \cdot \prod_{i \in D} h_i^{a_i})^{-c} \cdot d^{\hat{r}_3} \cdot h_0^{\hat{s}'} \cdot h^{-\hat{sk}} \cdot \prod_{j \in \bar{D}} h_j^{\hat{a}_j}$
5. $c'^* \leftarrow H(A', \bar{A}, d, nym, t_1^*, t_2^*, \hat{S}, g_1, pk, D, \text{attrs}_D)$
6. $c^* \leftarrow H(c'^*, m, n, bsn)$
7. If $c = c^*$ set $f \leftarrow 1$, otherwise $f \leftarrow 0$

VerifyLink : $(pk, \pi_\sigma, m, D, \text{attrs}_D, \pi_{\sigma^*}, m^*, D^*, \text{attrs}_{D^*}, bsn) \mapsto f$:

1. Extract nym from π_σ and nym^* from π_{σ^*}
2. $f_{\pi_\sigma} \leftarrow \text{VerifySPKSig}(D, \pi_\sigma, m, bsn, pk, \text{attrs}_D)$
3. $f_{\pi_{\sigma^*}} \leftarrow \text{VerifySPKSig}(D^*, \pi_{\sigma^*}, m^*, bsn, pk, \text{attrs}_{D^*})$
4. If $(f_{\pi_\sigma} = f_{\pi_{\sigma^*}} = 1) \wedge (nym = nym^*)$ set $f \leftarrow 1$ otherwise $f \leftarrow 0$

Figure 8: Specification of the verification algorithms of Π_{bbs+} .

3.3 Protocol Flows

Using the algorithms defined in Section 3.1 and Section 3.2, this section presents the high-level protocols of Π_{bbs+} . To be precise, this section specifies how ACs can be issued and how they can be presented to a verifier according to the requirements of minimality of information, unlinkability, non-duplication, and privacy-preserving revocation. As a disclaimer, due to the focus on the requirements above, technical details regarding the implementation of the protocols are beyond the scope of this paper. The first protocol in this section is **IssuerSetup**, which is specified in Figure 9.

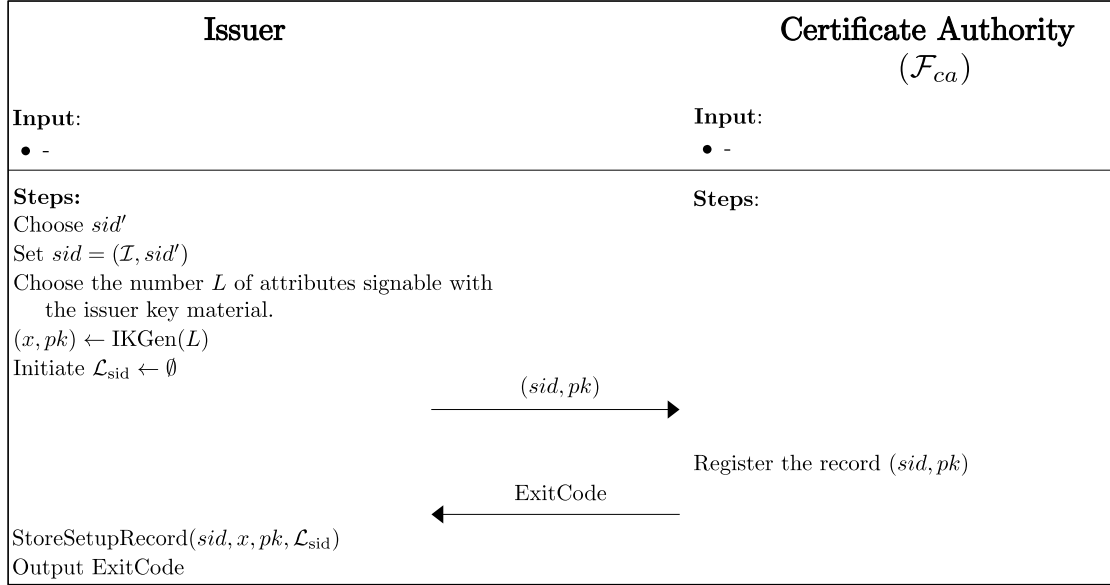


Figure 9: Specification of the **IssuerSetup** protocol.

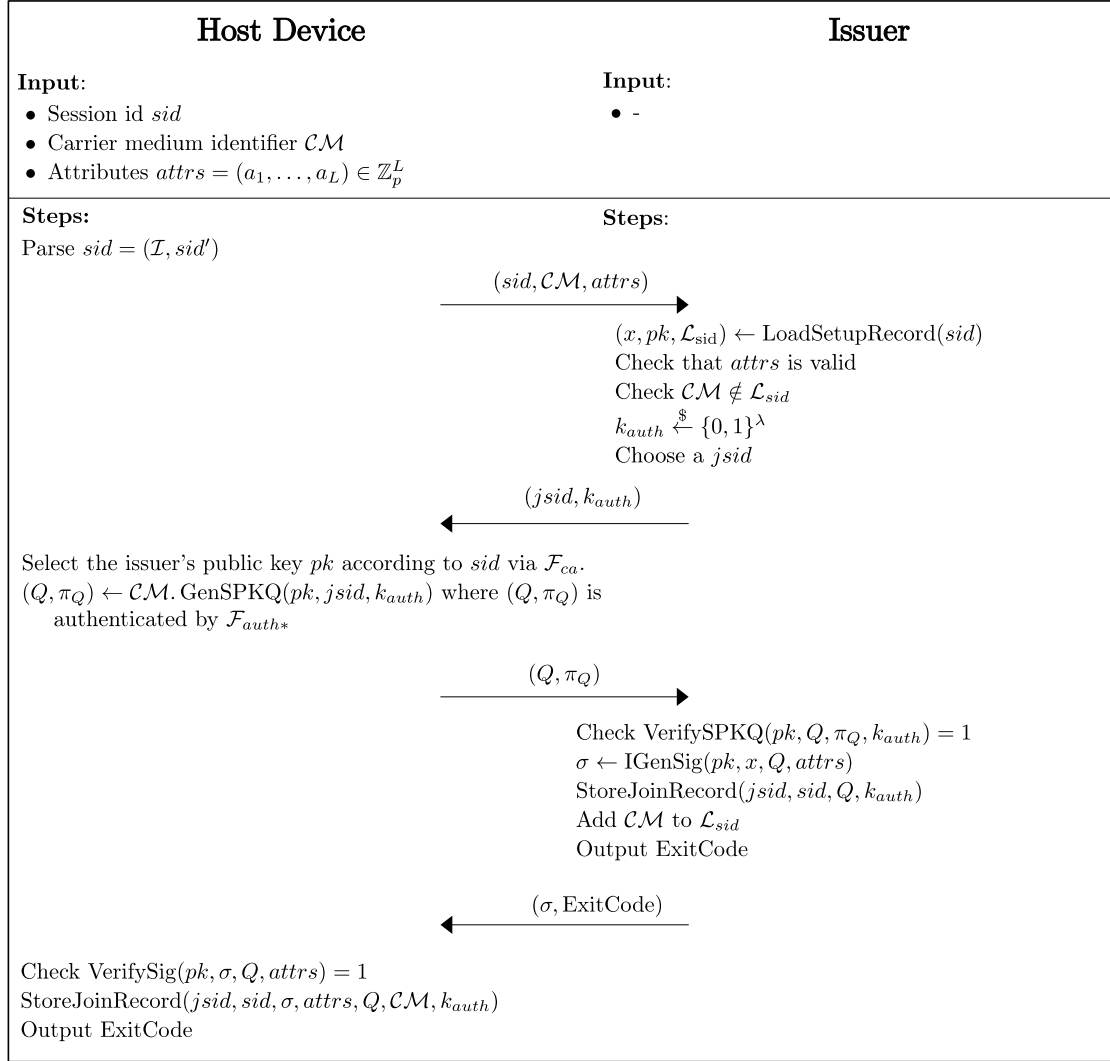
Besides generating the BBS+ key material (x, pk) , **IssuerSetup** sets the sid , which other protocols need to uniquely identify the generated key material. Furthermore, registering (sid, pk) at $\mathcal{F}_{ca}$ is particularly noteworthy, as it allows the other participants of Π_{bbs+} to access essential information such as the public key and the identifier sid of the issuer $\mathcal{I}$. Another task of **IssuerSetup** is to initialize $\mathcal{L}_{sid}$, which is the set of CM identifiers of host devices for which an issuer has issued an AC.

When an issuer has completed the **IssuerSetup**, everything is ready for the issuer to start issuing ACs. To issue an AC, the **Issuing** protocol, specified in Figure 10, must be executed between a host device and an issuer. For simplicity, in the **Issuing** protocol specification, the interaction between the holder and the host device is skipped, and the protocol starts directly on the host device for the input the holder desires.

In the **Issuing** protocol, the issuer inspects the potential AC attributes. The procedure by which the issuer must verify the accuracy of attributes is contingent upon the particular use case of the AC. Therefore, we cannot provide further details on attribute checking.

Upon successful execution, the host device and the issuer store a JoinRecord serving three purposes. First, it stores all the necessary data about an AC's issuance process. Second, an issuer uses it to decide whether the AC is valid. Third, the JoinRecord includes the key k_{auth} that serves as the key to the authentication mechanism used when issuing LVVCs.

Recall that, in the case of LVVC-based revocation, the issuer manages the revocation status of ACs. In detail, if an issuer has a JoinRecord for an AC, that specific AC is considered valid. Therefore, deleting a JoinRecord on the issuer's side revokes the corresponding AC. Hence, an

Figure 10: Specification of the **Issuing** protocol.

issuer can revoke an AC on its own. Invalidating an AC at the request of a holder requires an additional authentication procedure, which can be implemented in different ways depending on the use case and is, therefore, beyond the scope of this paper. How invalidating an AC leads to revocation is discussed after the specification of the **LVVCIssuing** protocol.

The **LVVCIssuing** protocol, specified in Figure 11, is a subprotocol that issues LVVCs for an AC. To verify whether a host device is eligible to obtain an LVVC for an AC, **LVVCIssuing** employs the standardized unilateral two-pass authentication mechanism of ISO/IEC 9798-2:2019 [ISO19]. The functions $\text{Enc}\{0, 1\}^\lambda \times \{0, 1\}^\lambda \rightarrow \{0, 1\}^\lambda$ and $\text{Dec}\{0, 1\}^\lambda \times \{0, 1\}^\lambda \rightarrow \{0, 1\}^\lambda$ represent the encryption and decryption functions of an authenticated encryption scheme, respectively. According to the current state of the art, e.g., the Advanced Encryption Standard (AES) [Nat01] operating in the Galois/Counter Mode (GCM) [Dwo07], is a suitable option. On successful authentication, the issuer generates an LVVC holding the revocation information for the corresponding AC.

In **LVVCIssuing**, selecting the revocation information of an LVVC is done using **GenRevocationInfo**. The concrete specification of **GenRevocationInfo** depends on the revocation information used, given by a concrete use case. Hence, in the **LVVCIssuing** protocol, **GenRevocationInfo** is ab-

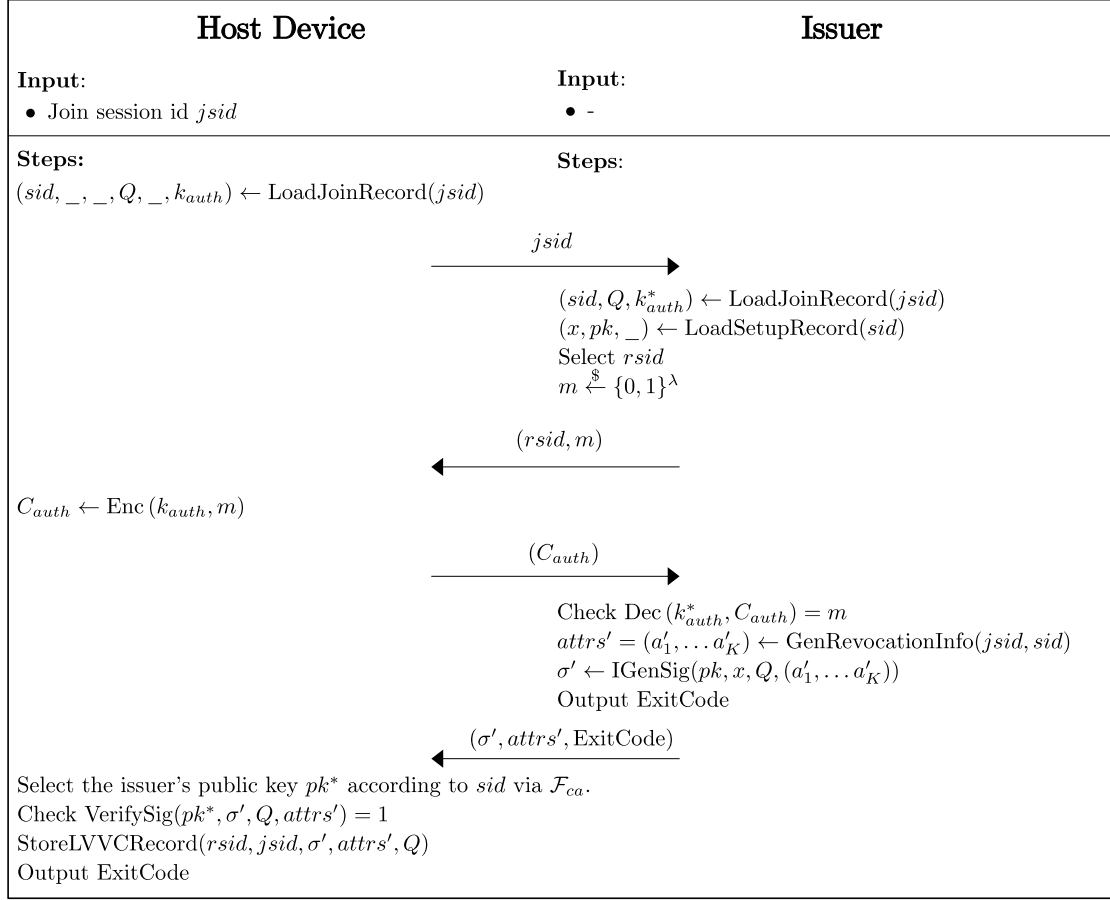


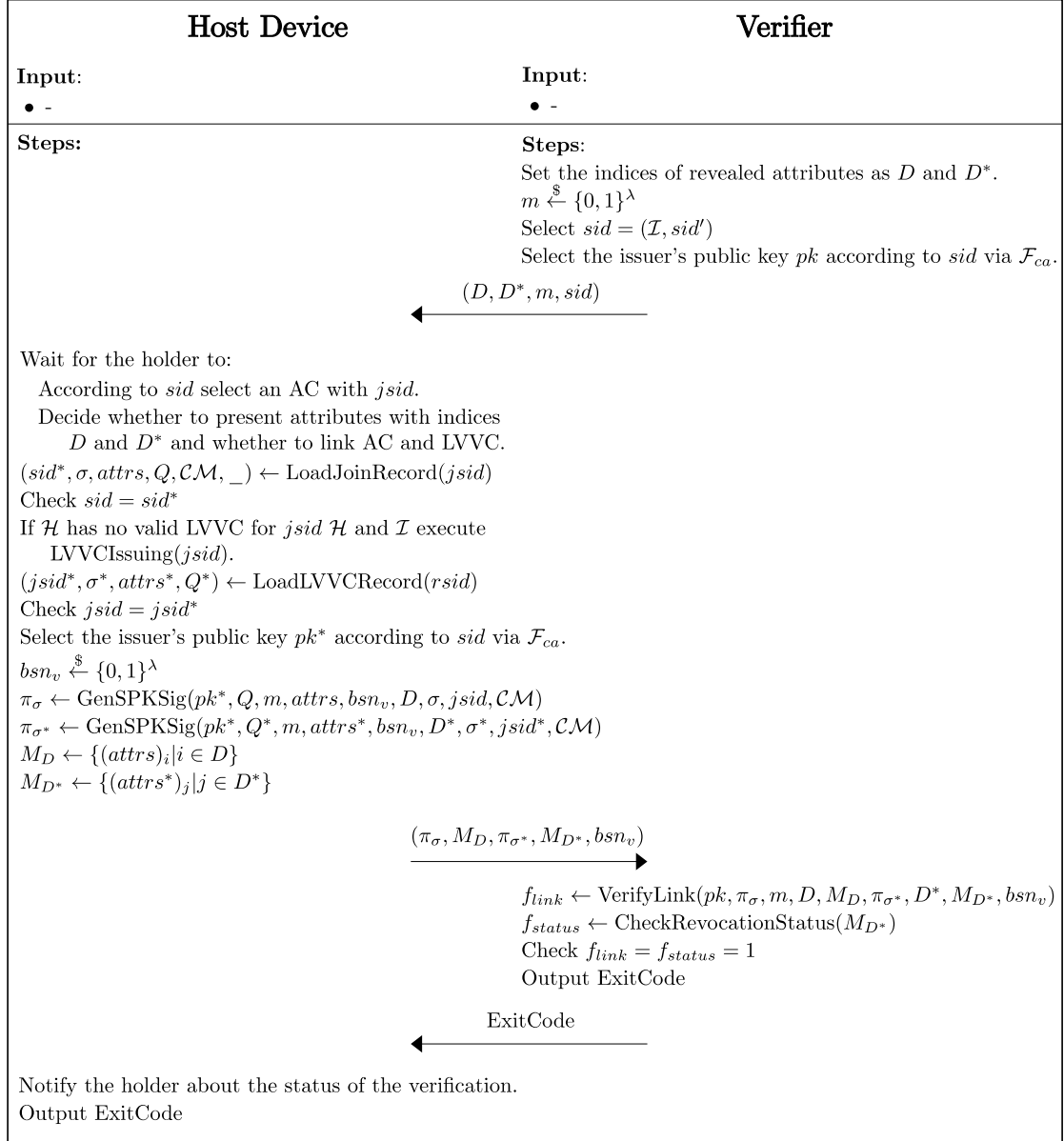
Figure 11: Specification of the **LVVCIssuing** protocol.

strictly defined. Here, on input $jsid$ and sid , **GenRevocationInfo** returns a K -tuple of attributes representing the revocation status.

A concrete form of revocation information proposed here is the date of issuance. For example, a verifier could only accept LVVCs with a specific issuance date, i.e., one that is at most one day in the past. Thus, LVVCs with time-based revocation information can expire. However, as long as an AC is valid (the issuer has a JoinRecord for it), a holder can use the **LVVCIssuing** protocol to obtain a new valid LVVC. With the issuance date as revocation information, an AC is revoked if a holder does not possess and cannot get an LVVC with an appropriate date.

As discussed in Section 2.3.1, using the issuance date as revocation information might only be an option for some use cases. Due to the unlinkability requirement, it is crucial to choose the issuance date so that it does not provide a unique identifier. See Section 3.4 for a discussion of a case where the issuance date complies with the unlinkability requirement.

The last protocol specified in this section is the **Presentation** protocol. As specified in Figure 12, the **Presentation** protocol proves the validity of an AC while exposing only attributes with the holder's consent. The holder's explicit consent and the selective disclosure of attributes keep the verification of an AC privacy-preserved. Furthermore, on the verifier site, the CheckRevocationStatus function returns 1 if the revocation information meets the security policies of the verifier. In addition, the **Presentation** protocol shows how the **VerifyLink** algorithm verifies the link between the VPs of an LVVC and an AC, proving the AC's revocation status to a verifier.

Figure 12: Specification of the **Presentation** protocol.

3.4 Security Discussion

This section discusses how the VPs used in Π_{bbs+} comply with the four requirements for minimality of information, unlinkability, non-duplication, and privacy-preserving revocation. The following discussion of the four mentioned requirements refers to VPs of an AC. We encourage the reader to recall the creation of VPs for an AC described in the **Presentation** protocol (see Figure 12).

Recall that during the **Presentation** protocol, the host device creates two VPs. In particular, VPs relate to an LVVC and an AC and are bundled as a tuple of information. To clarify, the tuple consists of disclosed attributes, a basename, and two SPKs generated by **GenSPKSig**.

After recalling the VPs of Π_{bbs+} , the subsequent analysis incorporates assumptions regarding Π_{bbs+} to facilitate a discussion on their security:

- The issuer is trustworthy.
- The issuer detects incorrect attributes based on the use case during the **Issuing** protocol.
- The ideal functionalities $\mathcal{F}_{ca}$, $\mathcal{F}_{crs}$, $\mathcal{F}_{smt}^l$, and $\mathcal{F}_{auth*}$ are trustworthy.
- The manufacturer of the CM is trustworthy.

Before discussing the requirements of minimality of information, unlinkability, non-duplication, and privacy-preserving revocation in detail, we address the security of the core/helper design. The core/helper approach is a trade-off between the number of computations made on the core device and protection against a malicious helper. Like other AC protocols [HS21, HSS23, CDL16a] using the core/helper approach, Π_{bbs+} cannot protect against malware on the helper, running sophisticated attacks against the holder's privacy or AC usage. Since Π_{bbs+} does not specify to encrypt attribute data, an attack against the holder's privacy by the helper, e.g., collecting information about the AC's attributes, is relatively straightforward and can be done without interacting with the core. In contrast, when exploiting an AC by manipulating or using its VPs, the helper must interact with the core device. Whether the mentioned risks are acceptable depends on the use case and must be considered when using Π_{bbs+} .

Minimality of Information The selective disclosure capabilities of SPKs with BBS+ signatures to create VPs fulfill the minimality of information requirement. In addition to selective disclosure, as specified in the **Presentation** protocol, before creating VPs, the holder must consent to disclose the attributes requested by a verifier. Hence, under the security guarantees of SPKs with BBS+ and with the holder's consent, the VPs in Π_{bbs+} only reveal the information necessary for verification.

Unlinkability Under the security guarantees of SPKs with BBS+ signatures, the holder can create VPs in an unlinkable fashion. Nevertheless, Π_{bbs+} uses LVVCs as its revocation mechanism, which uses intentional linking of the VPs of LVVC and AC. Linking the SPKs of LVVC and AC complies with the unlinkability requirement because, during the **Presentation** protocol, the holder has to consent to link an LVVC to an AC and disclose the requested attributes.

Before disclosing attributes, the holder must consider their uniqueness. Specifically, a disclosed attribute that displays a unique identifier leads to a linkable VP. In addition, combining a set of unlinkable attributes could also lead to linkable VPs, e.g., birthday, forename, surname, and address. Therefore, complying with unlinkability requires a holder to exclude disclosing attributes to multiple verifiers that form a unique combination.

In contrast to attributes, for each presentation session, the basename bsn_v that controls the link between two VPs must be unique. To prevent unintentional linking of VPs between presentation sessions, a holder must not use a basename bsn_v twice. If a holder uses a cryptographically secure random number generator with an appropriate security level λ , then the probability that a holder accidentally operates the same basename twice for different presentation sessions is negligible.

Non-Duplication Duplicating an LVVC or AC requires a malicious holder or attacker to copy all the data needed to create a valid SPK. Due to LVVC and AC sharing the same device secret, duplicating LVVC and AC is analogous. Therefore, the following discusses only the AC case.

The more secure the AC data are stored, the harder it is to duplicate. Specifically, in Π_{bbs+} , the CM stores the device secret, and the host device stores all other data about the AC. Since creating an SPK involves the device secret, an AC duplication attack must extract the device secret from the CM. Under the assumption that the CM's manufacturer is trustworthy, using a CM as device secret storage is a state-of-the-art measure to drastically minimize the duplication risk in most use cases. Consequently, a malicious holder or an attacker could copy all data except the device secret with a manageable effort.

Privacy-Preserving Revocation Fulfilling privacy-preserving revocation consists of two parts. First, the AC system requires a revocation mechanism. Second, the revocation mechanism does not violate the unlinkability requirement.

Before discussing the unlinkability, it is necessary to recall the revocation mechanism of Π_{bbs+} . Section 3.3 describes the use of LVVCs for revocation in Π_{bbs+} . To be precise, if a holder cannot present a valid LVVC for an AC, the AC is considered revoked.

Whether an LVVC is valid depends on the revocation information. In detail, Section 3.3 proposes using the LVVC issuance date as revocation information. When choosing specific values as revocation information, it is necessary to ensure that they adhere to the unlinkability requirement.

The way Π_{bbs+} complies with the unlinkability requirement is to prevent the revocation information from being a unique identifier. In particular, issuing LVVC for different holders with the same revocation information avoids the use of unique revocation information. Specifically, an issuer uses a fixed time frame, e.g., an hour, a day, or a week, as revocation information, which changes regularly. A verifier who receives an LVVC with a fixed time frame can now decide by its policies whether the AC is valid. Since the revocation information is not unique and Π_{bbs+} assumes that the issuer is trustworthy, without the holder's consent, a network of malicious verifiers cannot correlate VPs between presentation sessions.

4 Discussion

Interpretation of Results Designing an AC system that provides non-duplication and privacy-preserving revocation of ACs is particularly challenging because of the obligatory requirements of minimality of information and unlinkability. Some AC systems or similar protocols already implement non-duplication or revocation, and when they implement both, they often lack privacy [BHJ⁺10, CDL16a, HS21, BCGS09]. In this paper, we present the first specification of the privacy-by-design AC system Π_{bbs+} that satisfies the minimality of information, unlinkability, non-duplication, and privacy-preserving revocation utilizing the BBS+ signature scheme and LVVC.

For the specification of Π_{bbs+} , we transform the well-known DAA+ protocol by Camenisch et al. [CDL16a] into an AC system. Unlike the revocation list approach of DAA+, the Π_{bbs+} specification uses privacy-preserving revocation with LVVC. In particular, Π_{bbs+} uses a lightweight and straightforward method to implement privacy-preserving revocation.

Remarkably, the work of Hesse et al. [HSS23] and Camenisch et al. [CDL16a] shows that BBS+ signatures can be implemented using an SE or TPM as CM, respectively. In particular, the specification of Π_{bbs+} (see Section 3.2) uses SPKs similar to those of Hesse et al. [HSS23] and Camenisch et al. [CDL16a]. Therefore, it can be assumed that Π_{bbs+} can also be implemented with SE or TPM as CM.

Comparison Table 2 shows that, in comparison to the work of Hesse et al. [HSS23], Π_{bbs+} complies with the requirements of minimality of information, unlinkability, non-duplication, and privacy-preserving revocation. To the best of our knowledge, Π_{bbs+} is the only AC system that uses BBS+ that achieves all the criteria displayed in Table 2. Besides Π_{bbs+} , there are other AC systems that also satisfy all criteria. The specifications by Hanzlik and Slamanig [HS21] or Idemix [CL03, CL01, BC10, VA13] also fulfill the minimality of information, unlinkability, non-duplication, and privacy-preserving revocation. However, they use different procedures for the revocation of ACs, which have disadvantages in terms of their performance or complexity, as described in Section 2.3.

Table 2: Comparative evaluation of various AC system specifications including Π_{bbs+} .

Specification	Minimality of Information	Unlinkability	Non-Duplication	Revocation	Privacy-Preserving Revocation
Hanzlik & Slamanig [HS21]	AAEQ	AAEQ ZKPs	& Core/helper	Revocation list	Yes
Hesse et al. [HSS23]	BBS+	BBS+ ZKPs	& Core/helper	No	No
Bichsel et al. [BCGS09]	CL	CL & ZKPs	CM only	Revocation list	No
Batina et al. [BHJ ⁺ 10]	Self-blindable ECDSA	Self-blindable ECDSA & ZKPs	No	No	
U-Prove [PZ23, Paq23]	Blind digital signatures [ISO16]	No	CM only	Revocation list	No
Idemix [CL03, CL01, BC10]	CL	CL & ZKPs	CM only [VA13]	Cryptographic accumulator	Yes
Π_{bbs+} (specified in Section 3)	BBS+	BBS+ ZKP	& Core/helper	Time-based LVVCs	Yes

Implications The implications of Π_{bbs+} are substantial for the development of secure digital identity solutions using BBS+ signatures. By combining the core/helper approach with time-based LVVCs, Π_{bbs+} minimizes the risk of identity theft and can enhance user trust in digital identity frameworks. Thus, Π_{bbs+} is a viable candidate for future research in various applications requiring secure and anonymous authentication.

Limitations Despite the strengths of Π_{bbs+} , we acknowledge some limitations. In Section 3.4, we make several assumptions regarding Π_{bbs+} , and any compromise of these assumptions could undermine the integrity of the system. Besides that, Π_{bbs+} lacks a backup feature, and in case of a loss or nonfunctioning CM, the holder must perform the **Issuing** protocol again (using a new CM). Additionally, the time-based revocation mechanism may not be suitable for all applications, particularly those that require immediate revocation capabilities or very short validity periods of LVVCs. Another fact that limits the number of potential use cases for Π_{bbs+} is that the holder must be online during the issuance of LVVCs. Furthermore, even if the feasibility of Π_{bbs+} looks promising, there is still no implementation that proves its feasibility. Similarly, even though Section 3.4 conducts a security discussion, it does not give a rigorous security proof. However, the limitations regarding the implementation and rigorous security proof of Π_{bbs+} seem fixable, but we consider them outside the scope of this paper and as part of future work.

In the interpretation of the results, we assume that, similar to the work of Hesse et al. [HSS23] and Camenisch et al. [CDL16a], Π_{bbs+} can be implemented using an SE or a TPM, respectively. However, Hesse et al. [HSS23] only tested their implementation with NXP 1ID white plastic cards with a Smart MX D600 chip. Hence, the results of Hesse et al. [HSS23] cannot be generalized to all SEs. In addition, the feasibility of using a TPM as CM should also be examined. Thus,

checking the feasibility of Π_{bbs+} with an SE or a TPM as CM requires more extensive research, which we consider, however, to be outside the scope of this paper.

Future Research Directions Conducting a formal security proof of Π_{bbs+} could further strengthen its trust for future studies or standardization efforts. Moreover, evaluating its performance on commodity hardware would underline its feasibility. An empirical analysis on the feasibility of Π_{bbs+} would also allow a quantitative comparison with other implementations of AC systems.

Conclusion In this paper, we introduce a novel anonymous credential system, referred to as Π_{bbs+} , which effectively addresses the challenges of minimality of information, unlinkability, non-duplication, and privacy-preserving revocation. Compared to existing AC systems that utilize BBS+ signatures, Π_{bbs+} shows significant advantages, particularly by satisfying the requirements of minimality of information, unlinkability, non-duplication, and privacy-preserving revocation. Despite the promising results and theoretical feasibility of implementation using specialized hardware components such as SEs and TPMs, in this paper, we acknowledge several limitations, including the absence of practical implementation and rigorous security proof. Future research directions should focus on conducting a rigorous security proof and empirical evaluation of Π_{bbs+} to further validate its applicability and performance in real-world scenarios.

Declarations

Author Contributions: Conceptualization: C.G.; formal analysis: C.G.; funding acquisition: M.M.; investigation: C.G.; methodology: C.G.; project administration: C.G. and N.B.; supervision: C.G., N.B., and M.M.; validation: C.G., N.B., M.S., M.M.; visualization: C.G.; writing – original draft: C.G.; writing – review & editing: C.G., N.B., M.S., and M.M. All authors have read and agreed to the published version of the manuscript.

Funding: This research was funded by the Federal Ministry for Economic Affairs and Climate Action of Germany in the framework of ONCE grant number FKZ 01MN21003F.

Data Availability Statement: No datasets were created or analyzed during this research.

Acknowledgments: We acknowledge the use of Google Material Icons in the figures of this paper. These icons are licensed under the Apache License 2.0. Furthermore, the authors are grateful to the reviewers for their constructive feedback and insightful recommendations, which significantly improved the quality of the article.

Abbreviations

The following abbreviations are used in this manuscript:

AAEQ	Aggregatable Attribute-Based Equivalence Class Signature
AC	Anonymous Credential
AES	Advanced Encryption Standard
BBS	Boneh, Boyen, and Shacham
BG	Bilinear Group with Generators
CM	Carrier Medium
DDA	Direct Anonymous Attestation
ECDSA	Elliptic Curve Digital Signature Algorithm
GDPR	General Data Protection Regulation
LVVC	Linked Validity Verifiable Credential
SE	Secure Element
SoK	Systemization of Knowledge
SPK	Signatrue Proof of Knowledge
TPM	Trusted Platform Module
NFC	Near Field Communication
VC	Verifiable Credential
VCDM	Verifiable Vredentials Data Model
VP	Verifiable Presentation
ZKP	Zero Knowledge Proof

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